Supplementary Information

A microbiome-informed Bayesian adaptation framework for personalized nutrient profiling

All quantities below are participant-balanced unless stated otherwise, in units of mmol l^{-1} h. Grouped cross-validation uses family membership where available and participant identity otherwise, with a fixed seed. Confidence intervals are percentile intervals from 2,000 paired participant-bootstrap resamples. Permutations number 200 and random calibration draws number five. Sign tests exclude participants whose value is identical under both models being compared.

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Supplementary Note 1: Regression lock and two previously unreported facts

Eighteen quantities reported in the preceding version of this analysis were recomputed from the saved out-of-fold predictions. All reproduce within a tolerance of 1.5×10^{-3} (Supplementary Table S1).

Two facts follow from the same recomputation and were not previously reported. First, the personal response level accounts for 83.6% of the total improvement over the shared predictor, so a headline that quotes only the total conflates a level correction available to any method with the composition-dependent adaptation that is specific to this representation. Second, applying the conditional prior with no measured meals does not help. On the 902-meal query set the prior-only prediction gives 2.3917 against 2.3639 for the shared predictor; on the 594-meal common query set used for the calibration curve it gives 2.3610 against 2.3699. The direction depends on which later meals are scored, so biological context provides no dependable benefit before the first measured meal.

quantity	kind	published	recomputed	match
shared	rmse	2.3640	2.3639	True
k1	rmse	2.1720	2.1720	True
k3	rmse	1.9770	1.9772	True
k5	rmse	1.9610	1.9612	True
intercept_only	rmse	2.0410	2.0408	True
nutrients_only	rmse	2.1640	2.1644	True
intercept_mac	rmse	2.0100	2.0096	True
intercept_lipid	rmse	2.0520	2.0521	True
profile_k3	profile_rmse	1.8120	1.8123	True
profile_intercept_only	profile_rmse	1.8810	1.8813	True
profile_nutrients_only	profile_rmse	1.8530	1.8532	True
profile_intercept_mac	profile_rmse	1.8220	1.8220	True
profile_intercept_lipid	profile_rmse	1.8850	1.8852	True
profile_shared	profile_rmse	1.8810	1.8813	True
micro_vs_population_k3	difference	0.0061	0.0061	True
micro_vs_host_k3	difference	0.0042	0.0042	True
relative_shared_to_full	pct	16.4000	16.3560	True
relative_intercept_to_full	pct	3.1000	3.1140	True

Supplementary Note 2: Three-layer decomposition and the primary estimand

The primary estimand is the reduction in participant-balanced RMSE of the full three-coefficient model relative to intercept-only calibration at three calibration meals. The reduction relative to the shared predictor is reported for context only.

In the free-living cohort the shared predictor reaches 2.364, intercept-only calibration 2.041 and the full model 1.977 (Supplementary Table S2). The channel increment is 0.064 (95% CI, 0.022–0.107), a relative reduction of 3.11% (1.15–4.89), with lower error in 33 of 44 participants (sign test $P = 0.0013$). An unshrunk personal mean reaches 2.121, so the empirical-Bayes prior contributes 0.080 (95% CI, 0.016–0.154) on its own. The lipid channel is not supported in this cohort (0.032; 95% CI, −0.007 to 0.079; $P = 0.45$), whereas the carbohydrate channel is (0.075; 95% CI, 0.037–0.117; $P = 1.1 \times 10^{-4}$). Supplementary Table S3 gives every contrast on both metrics.

The bootstrap standard error of the primary estimand is 0.0216, giving a minimum detectable effect at 80% power of 0.0605 against an observed effect of 0.0636. This cohort is therefore only just powered for its own primary comparison and has no power at all for effects of order 0.006.

model	meals	people	rmse	profile rmse
full	902	44	1.9772	1.8123
intercept_plus_mac	902	44	2.0096	1.8220
intercept_only	902	44	2.0408	1.8813
intercept_plus_lipid	902	44	2.0521	1.8852
raw_personal_mean_k3	902	44	2.1211	1.8813
channels_without_intercept	902	44	2.1644	1.8532
shared	902	44	2.3639	1.8813

contrast	metric	method	reference	difference	diff lo	diff hi	relative pct	better	sign p
level_correction	rmse	2.0408	2.3639	0.3231	0.1562	0.4887	13.6677	30	0.0226
level_correction	profile_rmse	1.8813	1.8813	0.0000	-0.0000	0.0000	0.0000	14	0.0226
channel_increment_PRIMARY	rmse	1.9772	2.0408	0.0636	0.0225	0.1066	3.1142	33	0.0013
channel_increment_PRIMARY	profile_rmse	1.8123	1.8813	0.0690	0.0238	0.1192	3.6695	29	0.0488
total_gain_context	rmse	1.9772	2.3639	0.3866	0.2042	0.5639	16.3563	30	0.0226
total_gain_context	profile_rmse	1.8123	1.8813	0.0690	0.0238	0.1192	3.6695	29	0.0488
shrinkage_vs_raw_mean	rmse	2.0408	2.1211	0.0804	0.0160	0.1536	3.7881	30	0.0226
shrinkage_vs_raw_mean	profile_rmse	1.8813	1.8813	0.0000	-0.0000	0.0000	0.0000	11	0.0013
full_vs_raw_mean	rmse	1.9772	2.1211	0.1439	0.0717	0.2208	6.7843	36	0.0000
full_vs_raw_mean	profile_rmse	1.8123	1.8813	0.0690	0.0238	0.1192	3.6695	29	0.0488
mac_contribution	rmse	1.9772	2.0521	0.0748	0.0369	0.1165	3.6470	35	0.0001
mac_contribution	profile_rmse	1.8123	1.8852	0.0729	0.0273	0.1229	3.8667	29	0.0488
lipid_contribution	rmse	1.9772	2.0096	0.0324	-0.0072	0.0787	1.6105	25	0.4514
lipid_contribution	profile_rmse	1.8123	1.8220	0.0098	-0.0049	0.0254	0.5352	28	0.0961
intercept_necessity	rmse	1.9772	2.1644	0.1872	0.0706	0.3055	8.6471	27	0.1742
intercept_necessity	profile_rmse	1.8123	1.8532	0.0409	0.0104	0.0757	2.2072	28	0.0961

Supplementary Note 3: Prior arms, nested selection and expert review

Conditional priors are grouped into four arms that do not overlap: population, host, microbial and an exploratory host-plus-microbial arm. Selection runs separately inside each arm and the chosen configuration is asserted to remain in its arm. Without this discipline a microbial arm can be occupied by a host-plus-microbial representation, which makes any microbial increment uninterpretable. Supplementary Table S7 lists the configuration selected in every outer fold and arm; the microbial arm selected a taxon representation in four folds and a functional representation in one.

Six nutrition experts reviewed the nutrient-channel assignments and their supporting evidence in two rounds, before the adaptive analysis was run. In the second round 176 of 180 assignment judgements were rated reasonable and 169 of 180 evidence explanations were rated sufficient. A separate review of food-group explanations rated 68 of 72 reasonable. A third review of case communication rated 64 of 96 reasonable, that is 66.7% (95% CI, 56.8–75.3%), which is a negative finding and is reported as one.

fold	arm	config	inner rmse
0	population	masked__uniform__noise3.0	0.3001
0	host	masked__host__bw1.0__mean0.5	0.3008
0	micro	masked__taxa__bw0.3__mean0.5	0.3006
0	host__micro	masked__host__taxa__bw0.3__mean0.0	0.3008
1	population	masked__uniform__noise3.0	0.3027
1	host	masked__host__bw0.3__mean0.5	0.3007
1	micro	masked__taxa__bw1.0__mean0.5	0.3008
1	host__micro	masked__host__function__bw0.3__mean0.5	0.3007
2	population	masked__uniform__noise3.0	0.2916
2	host	masked__host__bw1.0__mean0.0	0.2920
2	micro	masked__taxa__bw0.3__mean0.0	0.2920
2	host__micro	masked__host__taxa__bw1.0__mean0.0	0.2920
3	population	masked__uniform__noise3.0	0.3044
3	host	masked__host__bw1.0__mean0.0	0.3040
3	micro	masked__taxa__bw0.3__mean0.5	0.3038
3	host__micro	masked__host__taxa__bw0.3__mean0.5	0.3039
4	population	masked__uniform__noise3.0	0.3008
4	host	masked__host__bw3.0__mean0.5	0.3007
4	micro	masked__function__bw3.0__mean0.0	0.3007
4	host__micro	masked__host__function__bw1.0__mean0.5	0.3007

Supplementary Note 4: Negative control for microbial conditioning

The microbiome enters the model in two roles: it shapes the prior estimated from training participants, and it selects which training participants a new participant resembles. The negative control breaks both. Every participant, in training and in held-out data, receives another participant's complete vector of 2,001 microbial and functional features. The empirical-Bayes moments and the conditional prior are then re-estimated and the evaluation repeated. Two hundred permutations were drawn.

The observed advantage of microbial conditioning over an unconditional prior is +0.00216. The permutation reference distribution has median +0.00229 and a 2.5th to 97.5th percentile range of 0.00189 to 0.00269, so the observed value sits at the 27.5th percentile with a permutation P of 0.73. The observed microbial-arm RMSE of 1.9812 is matched by a permuted median of 1.9811. Breaking the true microbiome-to-participant correspondence therefore changes nothing. What the conditional prior gains is attributable to weighting training participants at all, not to the biological variable used for the weighting.

metric	observed	perm median	perm q025	perm q975	perm quantile of observed	n perm at or above	permuta
micro_vs_population	0.00216	0.00229	0.00189	0.00269	0.27500	145	0.72637
micro_vs_host	-0.00183	-0.00170	-0.00211	-0.00130	0.27500	145	0.72637

Supplementary Note 5: External replication and frozen transport

The procedure was repeated in full, including nested selection, in 998 participants eating seven standardized meal compositions. Calibration meals were drawn at random because these meals carry no time order, and five independent draws were evaluated.

Beyond the personal offset the channels reduce error by 4.54% (95% CI, 3.71–5.39), with lower error in 615 of 998 participants. Both channels are supported, the lipid channel by 3.44% (95% CI, 0.0083–0.0131) and the carbohydrate channel by 1.21% (0.0029–0.0045). Against an unshrunk personal mean the full model gains 7.29%.

For transport, the shared predictor and the prior were fitted on the 998-participant cohort, frozen, and applied to a geographically separate cohort of 100 participants. No target-cohort data entered training or selection, and disjointness was asserted. The channel increment is 6.92% (95% CI, 0.0129–0.0327; $P = 0.005$). The personal response level does not transport (0.0010; 95% CI, −0.019 to 0.019; $P = 0.46$). Supplementary Tables S4 and S5 give all models and contrasts for both cohorts.

cohort	arm	model	meals	people	rmse	rmse sd across draws	profile rmse
PREDICT_UK	host	channels_only	3987.0000	998.0000	0.3342	0.0032	0.2674
PREDICT_UK	host	full	3987.0000	998.0000	0.3003	0.0033	0.2396
PREDICT_UK	host	intercept_only	3987.0000	998.0000	0.3148	0.0031	0.2462
PREDICT_UK	host	intercept_plus_lipid	3987.0000	998.0000	0.3040	0.0036	0.2417
PREDICT_UK	host	intercept_plus_mac	3987.0000	998.0000	0.3112	0.0028	0.2444
PREDICT_UK	host	raw_personal_mean	3987.0000	998.0000	0.3241	0.0033	0.2462
PREDICT_UK	host	shared	3987.0000	998.0000	0.3298	0.0034	0.2462
PREDICT_UK	host_micro	channels_only	3987.0000	998.0000	0.3340	0.0032	0.2673
PREDICT_UK	host_micro	full	3987.0000	998.0000	0.3003	0.0034	0.2396
PREDICT_UK	host_micro	intercept_only	3987.0000	998.0000	0.3148	0.0031	0.2462
PREDICT_UK	host_micro	intercept_plus_lipid	3987.0000	998.0000	0.3041	0.0036	0.2417
PREDICT_UK	host_micro	intercept_plus_mac	3987.0000	998.0000	0.3112	0.0028	0.2444
PREDICT_UK	host_micro	raw_personal_mean	3987.0000	998.0000	0.3241	0.0033	0.2462
PREDICT_UK	host_micro	shared	3987.0000	998.0000	0.3298	0.0034	0.2462
PREDICT_UK	micro	channels_only	3987.0000	998.0000	0.3340	0.0032	0.2673
PREDICT_UK	micro	full	3987.0000	998.0000	0.3004	0.0034	0.2396
PREDICT_UK	micro	intercept_only	3987.0000	998.0000	0.3147	0.0031	0.2462
PREDICT_UK	micro	intercept_plus_lipid	3987.0000	998.0000	0.3041	0.0037	0.2417
PREDICT_UK	micro	intercept_plus_mac	3987.0000	998.0000	0.3111	0.0028	0.2444
PREDICT_UK	micro	raw_personal_mean	3987.0000	998.0000	0.3241	0.0033	0.2462
PREDICT_UK	micro	shared	3987.0000	998.0000	0.3298	0.0034	0.2462
PREDICT_UK	population	channels_only	3987.0000	998.0000	0.3260	0.0029	0.2579
PREDICT_UK	population	full	3987.0000	998.0000	0.3023	0.0035	0.2422
PREDICT_UK	population	intercept_only	3987.0000	998.0000	0.3076	0.0032	0.2462
PREDICT_UK	population	intercept_plus_lipid	3987.0000	998.0000	0.3029	0.0036	0.2431
PREDICT_UK	population	intercept_plus_mac	3987.0000	998.0000	0.3050	0.0031	0.2448
PREDICT_UK	population	raw_personal_mean	3987.0000	998.0000	0.3241	0.0033	0.2462
PREDICT_UK	population	shared	3987.0000	998.0000	0.3298	0.0034	0.2462
PREDICT_US_transport	host	channels_only	399.0000	100.0000	0.3343	0.0091	0.2645
PREDICT_US_transport	host	full	399.0000	100.0000	0.3000	0.0052	0.2356
PREDICT_US_transport	host	intercept_only	399.0000	100.0000	0.3227	0.0050	0.2438
PREDICT_US_transport	host	intercept_plus_lipid	399.0000	100.0000	0.3051	0.0054	0.2380
PREDICT_US_transport	host	intercept_plus_mac	399.0000	100.0000	0.3184	0.0047	0.2424
PREDICT_US_transport	host	raw_personal_mean	399.0000	100.0000	0.3357	0.0059	0.2438
PREDICT_US_transport	host	shared	399.0000	100.0000	0.3231	0.0019	0.2438
PREDICT_US_transport	host_micro	channels_only	399.0000	100.0000	0.3341	0.0090	0.2643
PREDICT_US_transport	host_micro	full	399.0000	100.0000	0.3001	0.0052	0.2359
PREDICT_US_transport	host_micro	intercept_only	399.0000	100.0000	0.3223	0.0050	0.2438
PREDICT_US_transport	host_micro	intercept_plus_lipid	399.0000	100.0000	0.3051	0.0053	0.2382
PREDICT_US_transport	host_micro	intercept_plus_mac	399.0000	100.0000	0.3180	0.0047	0.2424
PREDICT_US_transport	host_micro	raw_personal_mean	399.0000	100.0000	0.3357	0.0059	0.2438
PREDICT_US_transport	host_micro	shared	399.0000	100.0000	0.3231	0.0019	0.2438
PREDICT_US_transport	micro	channels_only	399.0000	100.0000	0.3336	0.0088	0.2639
PREDICT_US_transport	micro	full	399.0000	100.0000	0.2998	0.0050	0.2357
PREDICT_US_transport	micro	intercept_only	399.0000	100.0000	0.3221	0.0050	0.2438
PREDICT_US_transport	micro	intercept_plus_lipid	399.0000	100.0000	0.3048	0.0053	0.2381
PREDICT_US_transport	micro	intercept_plus_mac	399.0000	100.0000	0.3179	0.0047	0.2424
PREDICT_US_transport	micro	raw_personal_mean	399.0000	100.0000	0.3357	0.0059	0.2438
PREDICT_US_transport	micro	shared	399.0000	100.0000	0.3231	0.0019	0.2438
PREDICT_US_transport	population	channels_only	399.0000	100.0000	0.3228	0.0052	0.2551
PREDICT_US_transport	population	full	399.0000	100.0000	0.2997	0.0039	0.2389
PREDICT_US_transport	population	intercept_only	399.0000	100.0000	0.3106	0.0040	0.2438
PREDICT_US_transport	population	intercept_plus_lipid	399.0000	100.0000	0.3013	0.0039	0.2400

cohort	arm	model	meals	people	rmse	rmse sd across draws	profile rmse
PREDICT_US_transport	population	intercept_plus_mac	399.0000	100.0000	0.3073	0.0038	0.2426
PREDICT_US_transport	population	raw_personal_mean	399.0000	100.0000	0.3357	0.0059	0.2438
PREDICT_US_transport	population	shared	399.0000	100.0000	0.3231	0.0019	0.2438

cohort	arm	contrast	metric	difference	diff lo	diff hi	relative pct	better	s
PREDICT_UK	host	channel_increment_PRIMARY	profile_rmse	0.0066	0.0049	0.0083	2.6859	577.8000	0
PREDICT_UK	host	channel_increment_PRIMARY	rmse	0.0146	0.0119	0.0172	4.6265	615.8000	0
PREDICT_UK	host	full_vs_raw_mean	profile_rmse	0.0066	0.0049	0.0083	2.6859	577.8000	0
PREDICT_UK	host	full_vs_raw_mean	rmse	0.0238	0.0198	0.0279	7.3481	617.0000	0
PREDICT_UK	host	level_correction	profile_rmse	0.0000	-0.0000	0.0000	0.0000	394.0000	0
PREDICT_UK	host	level_correction	rmse	0.0150	0.0095	0.0205	4.5487	596.4000	0
PREDICT_UK	host	lipid_contribution	profile_rmse	0.0049	0.0033	0.0065	1.9869	558.2000	0
PREDICT_UK	host	lipid_contribution	rmse	0.0109	0.0085	0.0133	3.5112	592.2000	0
PREDICT_UK	host	mac_contribution	profile_rmse	0.0021	0.0016	0.0026	0.8744	572.4000	0
PREDICT_UK	host	mac_contribution	rmse	0.0038	0.0030	0.0046	1.2415	579.2000	0
PREDICT_UK	host	shrinkage_vs_raw_mean	profile_rmse	-0.0000	-0.0000	0.0000	-0.0000	396.4000	0
PREDICT_UK	host	shrinkage_vs_raw_mean	rmse	0.0092	0.0076	0.0109	2.8546	576.4000	0
PREDICT_UK	host	total_gain_context	profile_rmse	0.0066	0.0049	0.0083	2.6859	577.8000	0
PREDICT_UK	host	total_gain_context	rmse	0.0296	0.0249	0.0342	8.9671	648.4000	0
PREDICT_UK	host_micro	channel_increment_PRIMARY	profile_rmse	0.0066	0.0049	0.0083	2.6691	577.2000	0
PREDICT_UK	host_micro	channel_increment_PRIMARY	rmse	0.0145	0.0118	0.0171	4.5961	612.4000	0
PREDICT_UK	host_micro	full_vs_raw_mean	profile_rmse	0.0066	0.0049	0.0083	2.6691	577.2000	0
PREDICT_UK	host_micro	full_vs_raw_mean	rmse	0.0238	0.0198	0.0278	7.3273	613.6000	0
PREDICT_UK	host_micro	level_correction	profile_rmse	0.0000	-0.0000	0.0000	0.0000	384.0000	0
PREDICT_UK	host_micro	level_correction	rmse	0.0150	0.0095	0.0205	4.5579	595.4000	0
PREDICT_UK	host_micro	lipid_contribution	profile_rmse	0.0048	0.0032	0.0064	1.9694	558.0000	0
PREDICT_UK	host_micro	lipid_contribution	rmse	0.0108	0.0084	0.0132	3.4829	589.8000	0
PREDICT_UK	host_micro	mac_contribution	profile_rmse	0.0021	0.0016	0.0026	0.8706	572.0000	0
PREDICT_UK	host_micro	mac_contribution	rmse	0.0037	0.0029	0.0046	1.2322	578.2000	0
PREDICT_UK	host_micro	shrinkage_vs_raw_mean	profile_rmse	0.0000	-0.0000	0.0000	0.0000	386.2000	0
PREDICT_UK	host_micro	shrinkage_vs_raw_mean	rmse	0.0093	0.0077	0.0109	2.8638	578.8000	0
PREDICT_UK	host_micro	total_gain_context	profile_rmse	0.0066	0.0049	0.0083	2.6691	577.2000	0
PREDICT_UK	host_micro	total_gain_context	rmse	0.0295	0.0248	0.0341	8.9468	648.4000	0
PREDICT_UK	micro	channel_increment_PRIMARY	profile_rmse	0.0065	0.0049	0.0083	2.6584	574.8000	0
PREDICT_UK	micro	channel_increment_PRIMARY	rmse	0.0143	0.0117	0.0169	4.5414	614.6000	0
PREDICT_UK	micro	full_vs_raw_mean	profile_rmse	0.0065	0.0049	0.0083	2.6584	574.8000	0
PREDICT_UK	micro	full_vs_raw_mean	rmse	0.0236	0.0197	0.0277	7.2890	614.0000	0
PREDICT_UK	micro	level_correction	profile_rmse	0.0000	-0.0000	0.0000	0.0000	378.2000	0
PREDICT_UK	micro	level_correction	rmse	0.0151	0.0096	0.0206	4.5732	595.4000	0
PREDICT_UK	micro	lipid_contribution	profile_rmse	0.0048	0.0032	0.0064	1.9648	559.6000	0
PREDICT_UK	micro	lipid_contribution	rmse	0.0107	0.0083	0.0131	3.4362	589.4000	0
PREDICT_UK	micro	mac_contribution	profile_rmse	0.0021	0.0016	0.0026	0.8604	575.6000	0
PREDICT_UK	micro	mac_contribution	rmse	0.0037	0.0029	0.0045	1.2077	581.4000	0
PREDICT_UK	micro	shrinkage_vs_raw_mean	profile_rmse	-0.0000	-0.0000	0.0000	-0.0000	390.2000	0
PREDICT_UK	micro	shrinkage_vs_raw_mean	rmse	0.0093	0.0077	0.0110	2.8793	578.8000	0
PREDICT_UK	micro	total_gain_context	profile_rmse	0.0065	0.0049	0.0083	2.6584	574.8000	0
PREDICT_UK	micro	total_gain_context	rmse	0.0294	0.0247	0.0340	8.9092	649.6000	0
PREDICT_UK	population	channel_increment_PRIMARY	profile_rmse	0.0039	0.0031	0.0047	1.6007	594.0000	0
PREDICT_UK	population	channel_increment_PRIMARY	rmse	0.0054	0.0037	0.0071	1.7443	546.8000	0
PREDICT_UK	population	full_vs_raw_mean	profile_rmse	0.0039	0.0031	0.0047	1.6007	594.0000	0
PREDICT_UK	population	full_vs_raw_mean	rmse	0.0218	0.0169	0.0269	6.7295	543.8000	0
PREDICT_UK	population	level_correction	profile_rmse	0.0000	-0.0000	0.0000	0.0000	381.4000	0
PREDICT_UK	population	level_correction	rmse	0.0222	0.0181	0.0262	6.7340	641.0000	0
PREDICT_UK	population	lipid_contribution	profile_rmse	0.0026	0.0018	0.0033	1.0549	563.8000	0
PREDICT_UK	population	lipid_contribution	rmse	0.0028	0.0014	0.0042	0.9160	524.2000	0
PREDICT_UK	population	mac_contribution	profile_rmse	0.0009	0.0006	0.0012	0.3701	552.6000	0
PREDICT_UK	population	mac_contribution	rmse	0.0006	0.0003	0.0010	0.2076	495.4000	0
PREDICT_UK	population	shrinkage_vs_raw_mean	profile_rmse	-0.0000	-0.0000	0.0000	-0.0000	395.6000	0
PREDICT_UK	population	shrinkage_vs_raw_mean	rmse	0.0164	0.0130	0.0200	5.0753	547.2000	0
PREDICT_UK	population	total_gain_context	profile_rmse	0.0039	0.0031	0.0047	1.6007	594.0000	0
PREDICT_UK	population	total_gain_context	rmse	0.0276	0.0245	0.0306	8.3620	695.4000	0
PREDICT_US_transport	host	channel_increment_PRIMARY	profile_rmse	0.0082	0.0027	0.0137	3.3586	57.8000	0
PREDICT_US_transport	host	channel_increment_PRIMARY	rmse	0.0227	0.0131	0.0335	7.0215	66.2000	0
PREDICT_US_transport	host	full_vs_raw_mean	profile_rmse	0.0082	0.0027	0.0137	3.3586	57.8000	0
PREDICT_US_transport	host	full_vs_raw_mean	rmse	0.0357	0.0211	0.0519	10.6236	66.2000	0
PREDICT_US_transport	host	level_correction	profile_rmse	0.0000	-0.0000	0.0000	0.0000	38.6000	0
PREDICT_US_transport	host	level_correction	rmse	0.0005	-0.0197	0.0191	0.1440	54.6000	0
PREDICT_US_transport	host	lipid_contribution	profile_rmse	0.0068	0.0017	0.0119	2.7925	60.2000	0
PREDICT_US_transport	host	lipid_contribution	rmse	0.0184	0.0100	0.0277	5.7887	65.2000	0

cohort	arm	contrast	metric	difference	diff lo	diff hi	relative pct	better	s
PREDICT_US_transport	host	mac_contribution	profile_rmse	0.0024	0.0003	0.0045	1.0128	57.4000	0
PREDICT_US_transport	host	mac_contribution	rmse	0.0051	0.0020	0.0085	1.6665	56.6000	0
PREDICT_US_transport	host	shrinkage_vs_raw_mean	profile_rmse	0.0000	-0.0000	0.0000	-0.0000	36.8000	0
PREDICT_US_transport	host	shrinkage_vs_raw_mean	rmse	0.0130	0.0074	0.0189	3.8753	61.8000	0
PREDICT_US_transport	host	total_gain_context	profile_rmse	0.0082	0.0027	0.0137	3.3586	57.8000	0
PREDICT_US_transport	host	total_gain_context	rmse	0.0231	0.0081	0.0378	7.1587	62.2000	0
PREDICT_US_transport	host_micro	channel_increment_PRIMARY	profile_rmse	0.0079	0.0025	0.0134	3.2415	58.0000	0
PREDICT_US_transport	host_micro	channel_increment_PRIMARY	rmse	0.0222	0.0127	0.0329	6.8839	65.6000	0
PREDICT_US_transport	host_micro	full_vs_raw_mean	profile_rmse	0.0079	0.0025	0.0134	3.2415	58.0000	0
PREDICT_US_transport	host_micro	full_vs_raw_mean	rmse	0.0356	0.0210	0.0520	10.5999	65.6000	0
PREDICT_US_transport	host_micro	level_correction	profile_rmse	0.0000	-0.0000	0.0000	-0.0000	40.2000	0
PREDICT_US_transport	host_micro	level_correction	rmse	0.0009	-0.0192	0.0193	0.2648	54.6000	0
PREDICT_US_transport	host_micro	lipid_contribution	profile_rmse	0.0065	0.0015	0.0116	2.6783	60.2000	0
PREDICT_US_transport	host_micro	lipid_contribution	rmse	0.0180	0.0097	0.0271	5.6447	64.6000	0
PREDICT_US_transport	host_micro	mac_contribution	profile_rmse	0.0023	0.0003	0.0045	0.9818	57.0000	0
PREDICT_US_transport	host_micro	mac_contribution	rmse	0.0050	0.0019	0.0084	1.6295	56.6000	0
PREDICT_US_transport	host_micro	shrinkage_vs_raw_mean	profile_rmse	-0.0000	-0.0000	0.0000	-0.0000	40.6000	0
PREDICT_US_transport	host_micro	shrinkage_vs_raw_mean	rmse	0.0134	0.0076	0.0195	3.9916	61.8000	0
PREDICT_US_transport	host_micro	total_gain_context	profile_rmse	0.0079	0.0025	0.0134	3.2415	58.0000	0
PREDICT_US_transport	host_micro	total_gain_context	rmse	0.0231	0.0083	0.0374	7.1338	62.2000	0
PREDICT_US_transport	micro	channel_increment_PRIMARY	profile_rmse	0.0080	0.0026	0.0135	3.3049	58.8000	0
PREDICT_US_transport	micro	channel_increment_PRIMARY	rmse	0.0223	0.0129	0.0327	6.9198	65.6000	0
PREDICT_US_transport	micro	full_vs_raw_mean	profile_rmse	0.0080	0.0026	0.0135	3.3049	58.8000	0
PREDICT_US_transport	micro	full_vs_raw_mean	rmse	0.0359	0.0213	0.0520	10.6799	67.0000	0
PREDICT_US_transport	micro	level_correction	profile_rmse	-0.0000	-0.0000	0.0000	-0.0000	40.4000	0
PREDICT_US_transport	micro	level_correction	rmse	0.0010	-0.0190	0.0194	0.3164	54.8000	0
PREDICT_US_transport	micro	lipid_contribution	profile_rmse	0.0066	0.0016	0.0118	2.7423	61.4000	0
PREDICT_US_transport	micro	lipid_contribution	rmse	0.0181	0.0098	0.0271	5.6815	65.4000	0
PREDICT_US_transport	micro	mac_contribution	profile_rmse	0.0024	0.0003	0.0044	0.9898	59.4000	0
PREDICT_US_transport	micro	mac_contribution	rmse	0.0050	0.0020	0.0083	1.6457	57.2000	0
PREDICT_US_transport	micro	shrinkage_vs_raw_mean	profile_rmse	-0.0000	-0.0000	0.0000	-0.0000	35.8000	0
PREDICT_US_transport	micro	shrinkage_vs_raw_mean	rmse	0.0136	0.0077	0.0197	4.0410	63.2000	0
PREDICT_US_transport	micro	total_gain_context	profile_rmse	0.0080	0.0026	0.0135	3.3049	58.8000	0
PREDICT_US_transport	micro	total_gain_context	rmse	0.0233	0.0084	0.0378	7.2179	61.0000	0
PREDICT_US_transport	population	channel_increment_PRIMARY	profile_rmse	0.0049	0.0024	0.0075	2.0097	60.0000	0
PREDICT_US_transport	population	channel_increment_PRIMARY	rmse	0.0109	0.0048	0.0177	3.4919	62.8000	0
PREDICT_US_transport	population	full_vs_raw_mean	profile_rmse	0.0049	0.0024	0.0075	2.0097	60.0000	0
PREDICT_US_transport	population	full_vs_raw_mean	rmse	0.0360	0.0181	0.0555	10.7026	59.6000	0
PREDICT_US_transport	population	level_correction	profile_rmse	-0.0000	-0.0000	0.0000	-0.0000	34.8000	0
PREDICT_US_transport	population	level_correction	rmse	0.0126	-0.0019	0.0258	3.8870	61.0000	0
PREDICT_US_transport	population	lipid_contribution	profile_rmse	0.0037	0.0013	0.0061	1.5146	60.8000	0
PREDICT_US_transport	population	lipid_contribution	rmse	0.0076	0.0027	0.0130	2.4597	60.8000	0
PREDICT_US_transport	population	mac_contribution	profile_rmse	0.0011	0.0001	0.0022	0.4560	57.4000	0
PREDICT_US_transport	population	mac_contribution	rmse	0.0016	0.0001	0.0032	0.5165	54.6000	0
PREDICT_US_transport	population	shrinkage_vs_raw_mean	profile_rmse	-0.0000	-0.0000	0.0000	-0.0000	36.6000	0
PREDICT_US_transport	population	shrinkage_vs_raw_mean	rmse	0.0251	0.0130	0.0379	7.4733	58.8000	0
PREDICT_US_transport	population	total_gain_context	profile_rmse	0.0049	0.0024	0.0075	2.0097	60.0000	0
PREDICT_US_transport	population	total_gain_context	rmse	0.0234	0.0131	0.0334	7.2453	67.0000	0

Supplementary Note 6: Prior-arm equivalence

The equivalence margin was fixed in advance at 0.005, one third of the channel increment. In the 998-participant cohort, host conditioning beats an unconditional prior by +0.00200 (95% CI, 0.00013–0.00380) and microbial conditioning by +0.00181 (−0.00006 to 0.00361). Microbial against host conditioning gives −0.00019 (−0.00058 to 0.00020), which falls inside the equivalence margin in all five draws. The bootstrap standard error implies a minimum detectable advantage at 80% power of 0.00056, which is 4% of the channel increment. The conclusion is equivalence with power, not an uninformative null.

cohort	arm	reference arm	difference	lo	hi	relative pct	better	sign p	draws equivalent
PREDICT_UK	host	population	0.00200	0.00013	0.00380	0.66081	529.80000	0.09434	5
PREDICT_UK	host_micro	host	-0.00007	-0.00032	0.00019	-0.02226	482.80000	0.35866	5
PREDICT_UK	host_micro	micro	0.00012	-0.00016	0.00042	0.04130	503.40000	0.62677	5
PREDICT_UK	micro	host	-0.00019	-0.00058	0.00020	-0.06359	494.60000	0.56817	5
PREDICT_UK	micro	population	0.00181	-0.00006	0.00361	0.59767	530.40000	0.11130	5
PREDICT_US_transport	host	population	-0.00028	-0.00593	0.00541	-0.08991	49.40000	0.43986	0
PREDICT_US_transport	host_micro	host	-0.00008	-0.00054	0.00036	-0.02685	48.60000	0.47924	5
PREDICT_US_transport	host_micro	micro	-0.00027	-0.00141	0.00081	-0.08990	51.20000	0.55087	5
PREDICT_US_transport	micro	host	0.00019	-0.00086	0.00133	0.06295	46.40000	0.59011	5
PREDICT_US_transport	micro	population	-0.00009	-0.00572	0.00547	-0.02660	48.60000	0.43184	0

Supplementary Note 7: Classical baselines

A linear mixed model with linear fixed effects, a random intercept and random channel slopes was fitted on the training participants. For each held-out participant the best linear unbiased predictor was computed from the identical calibration meals using the fitted variance components, so the comparison differs only in the model, not in the information supplied.

In free living the mixed model reaches 2.532 without calibration and 2.156 with it, against 1.980 for the full model, a margin of 8.16% (95% CI, 0.061–0.295; 31 of 44 participants). In standardized meals it reaches 0.369 and 0.357 against 0.299, a margin of 16.32% (95% CI, 0.0518–0.0647; 737 of 998). Decomposing the margin, the nonlinear shared predictor accounts for 10.76% in the larger cohort, where it is estimated precisely; in the 44-participant cohort the same comparison does not reach significance ($P = 0.291$). The personal calibration gain of the present method, 4.83%, exceeds that of the mixed model's own random slopes, 3.27%.

cohort	model	meals	people	rmse	profile rmse
CGMacros	shared	902	44	2.3639	1.8813
CGMacros	raw_personal_mean	902	44	2.1211	1.8813
CGMacros	intercept_only	902	44	2.0420	1.8813
CGMacros	full	902	44	1.9802	1.8152
CGMacros	lmm_population	902	44	2.5324	1.9956
CGMacros	lmm_blup	902	44	2.1561	1.9395
PREDICT_UK	shared	3987	998	0.3291	0.2482
PREDICT_UK	raw_personal_mean	3987	998	0.3232	0.2482
PREDICT_UK	intercept_only	3987	998	0.3137	0.2482
PREDICT_UK	full	3987	998	0.2985	0.2406
PREDICT_UK	lmm_population	3987	998	0.3688	0.2895
PREDICT_UK	lmm_blup	3987	998	0.3567	0.2841

cohort	contrast	method	reference	difference	diff lo	diff hi	relative pct	better	participants	si
CGMacros	gutnut_vs_lmm_blup	1.9802	2.1561	0.1759	0.0611	0.2953	8.1595	31	44	0.
CGMacros	gutnut_vs_lmm_population	1.9802	2.5324	0.5522	0.3421	0.7587	21.8064	35	44	0.
CGMacros	lmm_blup_vs_lmm_population	2.1561	2.5324	0.3763	0.1785	0.5918	14.8593	25	44	0.
CGMacros	shared_vs_lmm_population	2.3639	2.5324	0.1685	-0.0616	0.4265	6.6542	26	44	0.
CGMacros	gutnut_vs_intercept_only	1.9802	2.0420	0.0618	0.0192	0.1036	3.0262	33	44	0.
PREDICT_UK	gutnut_vs_lmm_blup	0.2985	0.3567	0.0582	0.0518	0.0647	16.3150	737	998	0.
PREDICT_UK	gutnut_vs_lmm_population	0.2985	0.3688	0.0703	0.0633	0.0773	19.0501	761	998	0.
PREDICT_UK	lmm_blup_vs_lmm_population	0.3567	0.3688	0.0120	0.0111	0.0130	3.2683	779	998	0.
PREDICT_UK	shared_vs_lmm_population	0.3291	0.3688	0.0397	0.0343	0.0451	10.7586	696	998	0.
PREDICT_UK	gutnut_vs_intercept_only	0.2985	0.3137	0.0152	0.0124	0.0179	4.8288	607	998	0.

Supplementary Note 8: Identifiability simulation

Personal coefficients were drawn from a known Gaussian population distribution with standard deviations $(1.2, 1.0, 1.0)$, and meals were simulated with correlated fibre and fat exposures mapped through the same saturating link. The grid crossed calibration length $k \in \{1, 3, 5, 10\}$, exposure correlation $\rho \in \{0, 0.3, 0.6, 0.9\}$, residual scale $\{0.5, 1, 2\}$ and prior displacement $\{0, 1\}$ prior standard deviations, with 400 replicates per cell.

With a correct prior, coverage at the 50%, 80% and 95% levels remains near nominal at every calibration length and correlation. Under a prior displaced by one prior standard deviation, nominal 95% coverage falls to 0.82–0.87. Coefficient recovery is poor throughout: at three meals the recovery error for a channel slope is 0.935 against a prior standard deviation of 1.0, and the posterior standard deviation shrinks by only 7.7%. Transfer error to unseen compositions nevertheless falls from 0.711 at one meal to 0.551 at three and 0.354 at ten. Three measured meals therefore locate a participant within the population distribution without estimating their coefficients, and personal adjustments must be read as model-based attributions.

k	rho	sigma	prior bias	rmse mac	rmse lipid	transfer rmse	cov50 mac	cov80 mac	cov95 mac	shrink mac	median cond
1	0.000	0.500	0.000	0.917	0.929	0.515	0.547	0.823	0.950	0.053	
3	0.000	0.500	0.000	0.819	0.878	0.351	0.510	0.802	0.963	0.143	26.733
5	0.000	0.500	0.000	0.803	0.741	0.291	0.525	0.785	0.945	0.215	12.145
10	0.000	0.500	0.000	0.662	0.618	0.221	0.477	0.795	0.945	0.344	9.024
1	0.000	0.500	1.000	1.319	1.339	0.628	0.343	0.635	0.855	0.051	
3	0.000	0.500	1.000	1.169	1.166	0.427	0.338	0.637	0.845	0.150	23.671
5	0.000	0.500	1.000	1.057	1.057	0.350	0.343	0.630	0.863	0.215	12.491
10	0.000	0.500	1.000	0.839	0.856	0.261	0.440	0.715	0.895	0.340	9.002
1	0.000	1.000	0.000	0.941	1.016	0.711	0.497	0.825	0.963	0.038	
3	0.000	1.000	0.000	0.935	0.913	0.550	0.522	0.797	0.943	0.077	25.065
5	0.000	1.000	0.000	0.859	0.911	0.456	0.517	0.805	0.965	0.102	12.726
10	0.000	1.000	0.000	0.853	0.874	0.354	0.470	0.777	0.955	0.164	9.056
1	0.000	1.000	1.000	1.385	1.323	0.964	0.333	0.605	0.810	0.036	
3	0.000	1.000	1.000	1.319	1.279	0.605	0.350	0.595	0.818	0.073	24.752
5	0.000	1.000	1.000	1.250	1.262	0.508	0.330	0.632	0.833	0.106	12.194
10	0.000	1.000	1.000	1.141	1.095	0.410	0.357	0.647	0.840	0.164	8.948
1	0.000	2.000	0.000	1.060	0.954	0.967	0.495	0.782	0.930	0.018	
3	0.000	2.000	0.000	0.937	1.011	0.773	0.485	0.805	0.963	0.039	25.164
5	0.000	2.000	0.000	0.919	0.951	0.680	0.530	0.785	0.960	0.053	12.093
10	0.000	2.000	0.000	0.866	0.958	0.532	0.517	0.843	0.963	0.078	9.059
1	0.000	2.000	1.000	1.394	1.358	1.201	0.307	0.600	0.845	0.019	
3	0.000	2.000	1.000	1.386	1.308	0.946	0.333	0.618	0.828	0.039	26.794
5	0.000	2.000	1.000	1.303	1.333	0.796	0.350	0.630	0.835	0.052	11.913
10	0.000	2.000	1.000	1.253	1.318	0.650	0.340	0.623	0.860	0.077	9.029
1	0.300	0.500	0.000	0.934	0.967	0.548	0.520	0.810	0.945	0.050	
3	0.300	0.500	0.000	0.915	0.870	0.359	0.463	0.765	0.932	0.140	27.122
5	0.300	0.500	0.000	0.752	0.796	0.285	0.485	0.795	0.960	0.216	12.062
10	0.300	0.500	0.000	0.590	0.646	0.217	0.550	0.848	0.965	0.343	8.819
1	0.300	0.500	1.000	1.349	1.350	0.614	0.347	0.618	0.795	0.052	
3	0.300	0.500	1.000	1.211	1.227	0.446	0.333	0.625	0.825	0.136	26.615
5	0.300	0.500	1.000	1.005	1.056	0.342	0.395	0.685	0.882	0.208	12.080
10	0.300	0.500	1.000	0.811	0.820	0.257	0.395	0.713	0.877	0.339	9.071
1	0.300	1.000	0.000	1.007	0.958	0.702	0.477	0.807	0.935	0.037	
3	0.300	1.000	0.000	0.894	0.888	0.494	0.525	0.833	0.945	0.077	25.405
5	0.300	1.000	0.000	0.872	0.846	0.441	0.497	0.818	0.955	0.104	12.499
10	0.300	1.000	0.000	0.870	0.849	0.342	0.485	0.780	0.930	0.167	8.957
1	0.300	1.000	1.000	1.308	1.346	0.912	0.357	0.625	0.843	0.035	
3	0.300	1.000	1.000	1.322	1.326	0.608	0.367	0.632	0.835	0.073	24.926
5	0.300	1.000	1.000	1.185	1.238	0.505	0.343	0.615	0.865	0.103	12.386
10	0.300	1.000	1.000	1.083	1.178	0.418	0.378	0.670	0.860	0.165	8.788
1	0.300	2.000	0.000	0.959	0.986	0.938	0.507	0.797	0.950	0.018	
3	0.300	2.000	0.000	0.965	0.955	0.766	0.537	0.800	0.940	0.041	25.401
5	0.300	2.000	0.000	0.965	0.952	0.627	0.492	0.760	0.955	0.052	12.073
10	0.300	2.000	0.000	0.937	0.962	0.558	0.482	0.767	0.948	0.077	9.004
1	0.300	2.000	1.000	1.356	1.375	1.180	0.320	0.650	0.848	0.017	
3	0.300	2.000	1.000	1.332	1.330	0.953	0.340	0.610	0.828	0.039	24.801
5	0.300	2.000	1.000	1.357	1.334	0.834	0.315	0.598	0.828	0.053	12.708
10	0.300	2.000	1.000	1.305	1.206	0.622	0.357	0.603	0.825	0.078	8.714
1	0.600	0.500	0.000	0.951	0.978	0.488	0.530	0.812	0.938	0.053	
3	0.600	0.500	0.000	0.861	0.842	0.335	0.512	0.795	0.950	0.143	26.011
5	0.600	0.500	0.000	0.764	0.825	0.290	0.517	0.807	0.953	0.209	12.397
10	0.600	0.500	0.000	0.683	0.690	0.224	0.527	0.785	0.927	0.337	8.818
1	0.600	0.500	1.000	1.317	1.274	0.626	0.380	0.625	0.835	0.052	
3	0.600	0.500	1.000	1.162	1.173	0.424	0.347	0.657	0.845	0.136	26.049
5	0.600	0.500	1.000	1.020	1.070	0.344	0.370	0.650	0.865	0.213	12.245

k	rho	sigma	prior bias	rmse mac	rmse lipid	transfer rmse	cov50 mac	cov80 mac	cov95 mac	shrink mac	median cond
10	0.600	0.500	1.000	0.819	0.853	0.262	0.390	0.688	0.905	0.334	9.057
1	0.600	1.000	0.000	0.899	0.978	0.720	0.573	0.843	0.955	0.034	
3	0.600	1.000	0.000	0.894	0.871	0.519	0.502	0.802	0.958	0.075	24.885
5	0.600	1.000	0.000	0.904	0.877	0.434	0.527	0.795	0.935	0.105	12.645
10	0.600	1.000	0.000	0.824	0.877	0.344	0.527	0.802	0.963	0.160	9.053
1	0.600	1.000	1.000	1.334	1.377	0.885	0.290	0.605	0.848	0.037	
3	0.600	1.000	1.000	1.312	1.299	0.631	0.360	0.598	0.815	0.073	24.788
5	0.600	1.000	1.000	1.253	1.309	0.534	0.347	0.630	0.833	0.102	12.212
10	0.600	1.000	1.000	1.125	1.094	0.409	0.335	0.645	0.858	0.159	9.022
1	0.600	2.000	0.000	1.030	1.009	0.927	0.468	0.782	0.943	0.018	
3	0.600	2.000	0.000	0.927	0.939	0.791	0.555	0.805	0.948	0.039	26.611
5	0.600	2.000	0.000	0.954	0.976	0.712	0.497	0.790	0.950	0.051	12.239
10	0.600	2.000	0.000	0.974	0.967	0.545	0.490	0.775	0.943	0.077	9.181
1	0.600	2.000	1.000	1.343	1.394	1.308	0.330	0.603	0.850	0.019	
3	0.600	2.000	1.000	1.253	1.410	0.901	0.323	0.655	0.870	0.039	25.663
5	0.600	2.000	1.000	1.384	1.360	0.808	0.287	0.600	0.802	0.052	12.381
10	0.600	2.000	1.000	1.266	1.282	0.617	0.340	0.620	0.820	0.076	8.917
1	0.900	0.500	0.000	0.919	0.980	0.481	0.480	0.828	0.968	0.048	
3	0.900	0.500	0.000	0.873	0.875	0.345	0.525	0.815	0.925	0.135	36.620
5	0.900	0.500	0.000	0.804	0.834	0.267	0.485	0.825	0.950	0.187	16.698
10	0.900	0.500	0.000	0.705	0.675	0.213	0.497	0.810	0.958	0.294	12.096
1	0.900	0.500	1.000	1.338	1.393	0.599	0.323	0.600	0.840	0.050	
3	0.900	0.500	1.000	1.173	1.179	0.383	0.335	0.632	0.858	0.135	38.810
5	0.900	0.500	1.000	1.135	1.050	0.332	0.347	0.647	0.825	0.188	16.585
10	0.900	0.500	1.000	0.860	0.925	0.243	0.390	0.682	0.895	0.296	11.882
1	0.900	1.000	0.000	0.979	0.967	0.731	0.470	0.785	0.950	0.036	
3	0.900	1.000	0.000	0.943	0.938	0.534	0.505	0.797	0.948	0.070	36.401
5	0.900	1.000	0.000	0.882	0.910	0.419	0.515	0.812	0.953	0.096	17.116
10	0.900	1.000	0.000	0.831	0.859	0.331	0.477	0.812	0.955	0.146	12.102
1	0.900	1.000	1.000	1.372	1.382	0.858	0.318	0.620	0.825	0.036	
3	0.900	1.000	1.000	1.353	1.246	0.597	0.333	0.585	0.825	0.070	33.489
5	0.900	1.000	1.000	1.283	1.282	0.506	0.333	0.623	0.830	0.098	17.132
10	0.900	1.000	1.000	1.183	1.173	0.398	0.355	0.625	0.855	0.146	12.558
1	0.900	2.000	0.000	0.952	0.992	0.931	0.497	0.818	0.955	0.019	
3	0.900	2.000	0.000	1.002	0.972	0.821	0.530	0.777	0.932	0.039	33.530
5	0.900	2.000	0.000	0.926	0.952	0.680	0.495	0.805	0.970	0.051	16.967
10	0.900	2.000	0.000	0.949	0.957	0.575	0.510	0.782	0.945	0.072	12.152
1	0.900	2.000	1.000	1.430	1.401	1.240	0.307	0.583	0.802	0.019	
3	0.900	2.000	1.000	1.395	1.414	0.946	0.318	0.585	0.800	0.038	35.455
5	0.900	2.000	1.000	1.289	1.371	0.791	0.343	0.620	0.840	0.050	16.170
10	0.900	2.000	1.000	1.329	1.267	0.671	0.335	0.598	0.828	0.073	12.352

Supplementary Note 9: Uncertainty calibration, design conditioning and a second sensor

The residual noise variance is a single empirical-Bayes scalar shared by all participants, taking values between 2.69 and 3.69 across outer folds. Three calibration meals and three coefficients leave no residual degree of freedom from which a per-participant value could be estimated.

Interval honesty is assessed by posterior predictive coverage rather than by a calibration error summary. Nominal 50%, 80% and 95% intervals cover 0.659, 0.871 and 0.963 of observations, and the standardized residuals have a standard deviation of 0.870 with a probability integral transform mean of 0.482. Intervals are therefore conservative. The calibration slope is 0.884 and the intercept 0.262, consistent with compression at high responses. Posterior standard deviations fall by 36.4%, 19.5% and 17.5% for the response level and the two channel slopes.

Three-meal calibration designs in free living are frequently ill conditioned. The median condition number is 31.3 and 39% of participants exceed 100. Prediction gain is nevertheless uncorrelated with conditioning, which is consistent with a prior-dominated posterior.

Each meal carries a second continuous glucose sensor on 1,093 of 1,122 occasions, correlated with the primary sensor at 0.865 with a median absolute difference of 0.679. This is a measurement replication and not a second biological endpoint. Repeating the analysis on the second sensor gives 2.637 for the shared predictor, 2.335 for intercept-only calibration and 2.243 for the full model, a channel increment of 0.0923 (95% CI, 0.0313–0.1566; 27 of 44; $P = 0.17$). The increment is reproduced in magnitude while remaining an order of magnitude smaller than the disagreement between the two sensors.

block	quantity	value
posterior_predictive_coverage	coverage_50	0.6585
posterior_predictive_coverage	coverage_50_participant_balanced	0.6588
posterior_predictive_coverage	coverage_80	0.8714
posterior_predictive_coverage	coverage_80_participant_balanced	0.8741
posterior_predictive_coverage	coverage_95	0.9634
posterior_predictive_coverage	coverage_95_participant_balanced	0.9643
calibration	calibration_slope	0.8841
calibration	calibration_intercept	0.2617
calibration	pit_mean	0.4824
calibration	pit_sd	0.2475
calibration	standardised_residual_sd	0.8702
calibration_design_conditioning	median	31.3147
calibration_design_conditioning	q25	15.8129
calibration_design_conditioning	q75	166075549731368608.0000
calibration_design_conditioning	q90	166075549731368608.0000
calibration_design_conditioning	max	1980955931924112896.0000
calibration_design_conditioning	share_above_100	0.3864
calibration_design_conditioning	gain_well_conditioned_median	0.0134
calibration_design_conditioning	gain_ill_conditioned_median	0.0365
calibration_design_conditioning	spearman_gain_vs_condition	-0.0026
posterior_contraction_pct	intercept	36.4500
posterior_contraction_pct	mac	19.5000
posterior_contraction_pct	lipid	17.5200
second_sensor_dexcom	meals	873.0000
second_sensor_dexcom	participants	44.0000
second_sensor_dexcom	shared	2.6365
second_sensor_dexcom	intercept_only	2.3351
second_sensor_dexcom	full	2.2428
second_sensor_dexcom_channel_increment	method	2.2428
second_sensor_dexcom_channel_increment	reference	2.3351
second_sensor_dexcom_channel_increment	difference	0.0923
second_sensor_dexcom_channel_increment	diff_lo	0.0313
second_sensor_dexcom_channel_increment	diff_hi	0.1566
second_sensor_dexcom_channel_increment	diff_se	0.0321
second_sensor_dexcom_channel_increment	relative_pct	3.9531
second_sensor_dexcom_channel_increment	rel_lo	1.4320
second_sensor_dexcom_channel_increment	rel_hi	6.3387
second_sensor_dexcom_channel_increment	participants	44.0000
second_sensor_dexcom_channel_increment	better	27.0000
second_sensor_dexcom_channel_increment	sign_p	0.1742
shared_residual_variance	fold_0_sigma2	3.2493
shared_residual_variance	fold_1_sigma2	3.3178
shared_residual_variance	fold_2_sigma2	3.4412
shared_residual_variance	fold_3_sigma2	2.6931

block	quantity	value
shared__residual__variance	fold__4__sigma2	3.6928

Supplementary Note 10: Cohort description, strata and responders

The free-living cohort comprises 14 participants with normal glucose tolerance, 16 with prediabetes and 14 with type 2 diabetes. Median age is 51 years, median body mass index 30.2 $\mathrm{kg\,m^{-2}}$, median HbA1c 5.95% and 28 of 44 participants are female. Of the meals, 6.6% have a positive incremental area of exactly zero. The standardized-meal cohorts have median ages of 44 and 41 years and median body mass indices of 25.3 and 25.9 $\mathrm{kg\,m^{-2}}$, with 28.6% of meals at exactly zero.

The channel increment is positive in all three glycaemic strata, with median per-participant gains of 0.028, 0.022 and 0.040 and 11 of 14, 12 of 16 and 10 of 14 participants improving.

group	participants	meals	age median	age iqr	bmi median	bmi iqr	hba1c median	hba1c iqr	fasting glucose medi
all	44	1122	51.00	43.75-58.25	30.16	26.94-35.93	5.95	5.50-6.90	110.50
healthy	14	365	40.00	28.50-48.50	26.46	23.42-30.72	5.35	5.12-5.47	97.00
prediabetes	16	410	54.00	45.75-58.25	32.76	28.83-36.22	5.95	5.80-6.12	110.50
t2d	14	347	52.50	51.00-59.00	30.62	28.24-34.35	7.10	6.93-7.35	149.00

cohort	participants	families	meals	meal compositions	age median	bmi median	female n	target median	target zer
PREDICT_UK	998	758	6981	7	44.00	25.30	651	0.54	28.63
PREDICT_US	100	100	699	7	41.00	25.90	68	0.57	27.61

Supplementary Note 11: Channel instantiation

The nutrient mask is a specification rather than a property of the method, so the choice of exposures is an empirical question. Six instantiations were compared in the 998-participant cohort with saturation constants set to the median non-zero exposure within each training fold.

Total fat is the best lipid coordinate of those tested. Full-model error is 0.3077 with a twelve-item carbohydrate composite and total fat, and 0.3079 with fibre and total fat. Replacing total fat by a fatty-acid fraction is worse in every case, reaching 0.3410 for saturated fat. The channel increment itself becomes negative for the saturated and polyunsaturated instantiations, meaning that personal calibration on those coordinates actively harms prediction. A composite of twelve expert-assigned carbohydrate exposures performs equivalently to fibre alone.

The channel coordinate therefore needs to carry dose rather than compositional quality. This conclusion is bounded by the composition space available: across only seven standardized meals, fatty-acid fractions are nearly collinear with total fat and with meal identity.

variant	model	meals	people	rmse	rmse sd	profile rmse
fibre_mufa	full	3987.0000	998.0000	0.3140	0.0063	0.2447
fibre_mufa	intercept_only	3987.0000	998.0000	0.3198	0.0070	0.2459
fibre_mufa	intercept_plus_lipid	3987.0000	998.0000	0.3099	0.0065	0.2405
fibre_mufa	intercept_plus_mac	3987.0000	998.0000	0.3160	0.0066	0.2454
fibre_mufa	shared	3987.0000	998.0000	0.3343	0.0046	0.2459
fibre_pufa	full	3987.0000	998.0000	0.3276	0.0051	0.2578
fibre_pufa	intercept_only	3987.0000	998.0000	0.3169	0.0070	0.2459
fibre_pufa	intercept_plus_lipid	3987.0000	998.0000	0.3222	0.0051	0.2534
fibre_pufa	intercept_plus_mac	3987.0000	998.0000	0.3131	0.0067	0.2455
fibre_pufa	shared	3987.0000	998.0000	0.3343	0.0046	0.2459
fibre_sfa	full	3987.0000	998.0000	0.3409	0.0081	0.2691
fibre_sfa	intercept_only	3987.0000	998.0000	0.3157	0.0069	0.2459
fibre_sfa	intercept_plus_lipid	3987.0000	998.0000	0.3360	0.0081	0.2643
fibre_sfa	intercept_plus_mac	3987.0000	998.0000	0.3129	0.0067	0.2455
fibre_sfa	shared	3987.0000	998.0000	0.3343	0.0046	0.2459
fibre_totalfat	full	3987.0000	998.0000	0.3079	0.0065	0.2410
fibre_totalfat	intercept_only	3987.0000	998.0000	0.3189	0.0070	0.2459
fibre_totalfat	intercept_plus_lipid	3987.0000	998.0000	0.3088	0.0065	0.2404
fibre_totalfat	intercept_plus_mac	3987.0000	998.0000	0.3159	0.0067	0.2452
fibre_totalfat	shared	3987.0000	998.0000	0.3343	0.0046	0.2459
macmix_sfa	full	3987.0000	998.0000	0.3276	0.0075	0.2580
macmix_sfa	intercept_only	3987.0000	998.0000	0.3160	0.0069	0.2460
macmix_sfa	intercept_plus_lipid	3987.0000	998.0000	0.3246	0.0075	0.2545
macmix_sfa	intercept_plus_mac	3987.0000	998.0000	0.3127	0.0069	0.2453
macmix_sfa	shared	3987.0000	998.0000	0.3343	0.0046	0.2460
macmix_totalfat	full	3987.0000	998.0000	0.3077	0.0065	0.2408
macmix_totalfat	intercept_only	3987.0000	998.0000	0.3181	0.0070	0.2460
macmix_totalfat	intercept_plus_lipid	3987.0000	998.0000	0.3093	0.0065	0.2412
macmix_totalfat	intercept_plus_mac	3987.0000	998.0000	0.3166	0.0070	0.2457
macmix_totalfat	shared	3987.0000	998.0000	0.3345	0.0046	0.2460

variant	contrast	difference	lo	hi	relative pct	better	sign p	draws
fibre_mufa	channel_increment	0.0058	0.0022	0.0095	1.8092	510.0000	0.5287	3
fibre_mufa	lipid_contribution	0.0021	-0.0012	0.0054	0.6534	502.6667	0.8421	3
fibre_mufa	mac_contribution	-0.0041	-0.0058	-0.0024	-1.3186	440.0000	0.0039	3
fibre_pufa	channel_increment	-0.0107	-0.0144	-0.0069	-3.3727	422.3333	0.0000	3
fibre_pufa	lipid_contribution	-0.0145	-0.0178	-0.0112	-4.6360	400.6667	0.0000	3
fibre_pufa	mac_contribution	-0.0054	-0.0069	-0.0039	-1.6669	425.6667	0.0009	3
fibre_sfa	channel_increment	-0.0252	-0.0291	-0.0215	-7.9756	336.0000	0.0000	3
fibre_sfa	lipid_contribution	-0.0280	-0.0319	-0.0244	-8.9647	310.3333	0.0000	3
fibre_sfa	mac_contribution	-0.0049	-0.0059	-0.0040	-1.4708	388.3333	0.0000	3
fibre_totalfat	channel_increment	0.0110	0.0084	0.0136	3.4412	569.0000	0.0001	3
fibre_totalfat	lipid_contribution	0.0079	0.0058	0.0102	2.5143	561.3333	0.0001	3
fibre_totalfat	mac_contribution	0.0009	-0.0001	0.0019	0.2871	511.0000	0.1744	3
macmix_sfa	channel_increment	-0.0115	-0.0142	-0.0089	-3.6415	392.3333	0.0000	3
macmix_sfa	lipid_contribution	-0.0148	-0.0172	-0.0126	-4.7382	343.6667	0.0000	3
macmix_sfa	mac_contribution	-0.0029	-0.0038	-0.0020	-0.9005	404.0000	0.0000	3
macmix_totalfat	channel_increment	0.0104	0.0083	0.0127	3.2764	590.3333	0.0000	3
macmix_totalfat	lipid_contribution	0.0089	0.0070	0.0109	2.8120	596.6667	0.0000	3
macmix_totalfat	mac_contribution	0.0016	0.0011	0.0022	0.5257	560.0000	0.0029	3

fold	variant	u channel1	u channel2	channel1	channel2
0	fibre_totalfat	1.850	22.200	exposure___Fiber, total dietary (g)	exposure___Total Fat (g)
0	fibre_sfa	1.850	3.496	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total saturated (g)
0	fibre_mufa	1.850	9.500	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total monounsaturated (g)
0	fibre_pufa	1.850	7.315	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total polyunsaturated (g)
0	macmix_totalfat	8.716	22.200	___macmix___	exposure___Total Fat (g)
0	macmix_sfa	8.716	3.496	___macmix___	exposure___Fatty acids, total saturated (g)
1	fibre_totalfat	1.700	22.200	exposure___Fiber, total dietary (g)	exposure___Total Fat (g)
1	fibre_sfa	1.700	3.496	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total saturated (g)
1	fibre_mufa	1.700	9.500	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total monounsaturated (g)
1	fibre_pufa	1.700	7.315	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total polyunsaturated (g)
1	macmix_totalfat	8.717	22.200	___macmix___	exposure___Total Fat (g)
1	macmix_sfa	8.717	3.496	___macmix___	exposure___Fatty acids, total saturated (g)
2	fibre_totalfat	1.850	22.200	exposure___Fiber, total dietary (g)	exposure___Total Fat (g)
2	fibre_sfa	1.850	3.496	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total saturated (g)
2	fibre_mufa	1.850	9.500	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total monounsaturated (g)
2	fibre_pufa	1.850	7.315	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total polyunsaturated (g)
2	macmix_totalfat	8.716	22.200	___macmix___	exposure___Total Fat (g)
2	macmix_sfa	8.716	3.496	___macmix___	exposure___Fatty acids, total saturated (g)
3	fibre_totalfat	1.850	22.200	exposure___Fiber, total dietary (g)	exposure___Total Fat (g)
3	fibre_sfa	1.850	3.496	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total saturated (g)
3	fibre_mufa	1.850	9.500	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total monounsaturated (g)
3	fibre_pufa	1.850	7.315	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total polyunsaturated (g)
3	macmix_totalfat	8.716	22.200	___macmix___	exposure___Total Fat (g)
3	macmix_sfa	8.716	3.496	___macmix___	exposure___Fatty acids, total saturated (g)
4	fibre_totalfat	2.000	22.200	exposure___Fiber, total dietary (g)	exposure___Total Fat (g)
4	fibre_sfa	2.000	3.496	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total saturated (g)
4	fibre_mufa	2.000	9.500	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total monounsaturated (g)
4	fibre_pufa	2.000	7.315	exposure___Fiber, total dietary (g)	exposure___Fatty acids, total polyunsaturated (g)
4	macmix_totalfat	8.716	22.200	___macmix___	exposure___Total Fat (g)
4	macmix_sfa	8.716	3.496	___macmix___	exposure___Fatty acids, total saturated (g)

Supplementary Note 12: Evaluation protocols and the calibration curve

Three protocols were compared. Under the fixed protocol every history length is scored on one common query set. Under the deployment protocol the query set begins immediately after calibration. Under the random protocol calibration meals are drawn at random rather than taken as the first k.

The deployment protocol, which is the realistic one and scores more meals, gives a larger channel increment than the fixed protocol (0.0736 against 0.0618 at three meals), so the fixed design was not favourable. The random protocol, however, reduces the three-meal increment to 0.0343 with an interval crossing zero ($P = 0.55$); significance returns at five meals. Read together with the condition numbers of Note 9, this identifies a requirement on the calibration set rather than a failure of the method: measured meals must span the channel exposures. In standardized meals, which span a wide composition range by design, the increment is strong under random calibration.

The calibration curve was extended to eleven meals so that the point at which it flattens is observed. On the common query set, full-model error is 2.361 with the prior alone, 2.143 after one meal, 2.053 after two, 1.936 after three, 1.919 after five, 1.891 after seven, 1.918 after nine and 1.886 after eleven. Three meals capture approximately nine tenths of the improvement that eleven meals reach. The channel increment is present at every history length, between 3.3% and 5.5% on the common query set.

protocol	history k	model	meals	people	rmse	profile rmse
deployment	1	full	1078.0000	44.0000	2.1421	1.8045
deployment	1	intercept_only	1078.0000	44.0000	2.2330	1.8633
deployment	1	raw_personal_mean	1078.0000	44.0000	3.0513	1.8633
deployment	1	shared	1078.0000	44.0000	2.3350	1.8633
deployment	3	full	990.0000	44.0000	1.9530	1.7984
deployment	3	intercept_only	990.0000	44.0000	2.0266	1.8736
deployment	3	raw_personal_mean	990.0000	44.0000	2.1050	1.8736
deployment	3	shared	990.0000	44.0000	2.3445	1.8736
deployment	5	full	902.0000	44.0000	1.9625	1.8064
deployment	5	intercept_only	902.0000	44.0000	2.0156	1.8813
deployment	5	raw_personal_mean	902.0000	44.0000	2.0537	1.8813
deployment	5	shared	902.0000	44.0000	2.3639	1.8813
fixed	1	full	902.0000	44.0000	2.1681	1.8266
fixed	1	intercept_only	902.0000	44.0000	2.2616	1.8813
fixed	1	raw_personal_mean	902.0000	44.0000	3.0687	1.8813
fixed	1	shared	902.0000	44.0000	2.3639	1.8813
fixed	3	full	902.0000	44.0000	1.9802	1.8152
fixed	3	intercept_only	902.0000	44.0000	2.0420	1.8813
fixed	3	raw_personal_mean	902.0000	44.0000	2.1211	1.8813
fixed	3	shared	902.0000	44.0000	2.3639	1.8813
fixed	5	full	902.0000	44.0000	1.9625	1.8064
fixed	5	intercept_only	902.0000	44.0000	2.0156	1.8813
fixed	5	raw_personal_mean	902.0000	44.0000	2.0537	1.8813
fixed	5	shared	902.0000	44.0000	2.3639	1.8813
random	1	full	902.0000	44.0000	2.1961	1.8659
random	1	intercept_only	902.0000	44.0000	2.2142	1.8810
random	1	raw_personal_mean	902.0000	44.0000	2.6868	1.8810
random	1	shared	902.0000	44.0000	2.3447	1.8810
random	3	full	902.0000	44.0000	2.0490	1.8458
random	3	intercept_only	902.0000	44.0000	2.0833	1.8810
random	3	raw_personal_mean	902.0000	44.0000	2.2250	1.8810
random	3	shared	902.0000	44.0000	2.3447	1.8810
random	5	full	902.0000	44.0000	1.9936	1.8309
random	5	intercept_only	902.0000	44.0000	2.0490	1.8810
random	5	raw_personal_mean	902.0000	44.0000	2.1189	1.8810
random	5	shared	902.0000	44.0000	2.3447	1.8810

protocol	history k	difference	lo	hi	relative pct	better	sign p	draws
deployment	1	0.0909	0.0339	0.1483	4.0690	27.0000	0.1742	1
deployment	3	0.0736	0.0289	0.1160	3.6318	33.0000	0.0013	1
deployment	5	0.0531	0.0127	0.0964	2.6343	29.0000	0.0488	1
fixed	1	0.0935	0.0298	0.1579	4.1343	24.0000	0.6516	1
fixed	3	0.0618	0.0192	0.1036	3.0262	33.0000	0.0013	1
fixed	5	0.0531	0.0127	0.0964	2.6343	29.0000	0.0488	1
random	1	0.0181	-0.0179	0.0590	0.8099	23.2000	0.7327	5

protocol	history k	difference	lo	hi	relative pct	better	sign p	draws
random	3	0.0343	-0.0077	0.0765	1.6502	24.8000	0.5462	5
random	5	0.0554	0.0055	0.1059	2.6927	25.4000	0.4348	5

protocol	history k	profile rmse	profile rmse.1	profile rmse.2	rmse	rmse.1	rmse.2
nan		full	intercept_only	shared	full	intercept_only	shared
common	0.0000	1.8478	1.8526	1.8526	2.361	2.3699	2.3699
common	1.0000	1.7794	1.8526	1.8526	2.1427	2.2676	2.3699
common	2.0000	1.7778	1.8526	1.8526	2.0528	2.1232	2.3699
common	3.0000	1.757	1.8526	1.8526	1.9355	2.029	2.3699
common	5.0000	1.7399	1.8526	1.8526	1.9186	2.0026	2.3699
common	7.0000	1.7428	1.8526	1.8526	1.8909	1.9877	2.3699
common	9.0000	1.7447	1.8526	1.8526	1.9184	2.0081	2.3699
common	11.0000	1.755	1.8526	1.8526	1.8863	1.9774	2.3699
deployment	0.0000	1.8598	1.8633	1.8633	2.3296	2.335	2.335
deployment	1.0000	1.8045	1.8633	1.8633	2.1421	2.2329	2.335
deployment	2.0000	1.8105	1.8733	1.8733	2.0449	2.1026	2.3522
deployment	3.0000	1.7984	1.8736	1.8736	1.953	2.0266	2.3445
deployment	5.0000	1.8064	1.8813	1.8813	1.9625	2.0156	2.3639
deployment	7.0000	1.7807	1.8555	1.8555	1.9209	1.9869	2.3384
deployment	9.0000	1.7341	1.818	1.818	1.9386	1.9994	2.3574
deployment	11.0000	1.7408	1.834	1.834	1.8866	1.9699	2.3685

protocol	history k	contrast	difference	diff lo	diff hi	relative pct	better	sign p
common	1	channel_increment	0.1249	0.0397	0.2164	5.5078	32	0.0037
common	1	total_vs_shared	0.2272	0.0246	0.4029	9.5854	24	0.6516
common	2	channel_increment	0.0704	0.0108	0.1268	3.3155	30	0.0226
common	2	total_vs_shared	0.3171	0.1410	0.4882	13.3805	25	0.4514
common	3	channel_increment	0.0935	0.0282	0.1572	4.6091	31	0.0096
common	3	total_vs_shared	0.4344	0.2465	0.6049	18.3301	26	0.2912
common	5	channel_increment	0.0839	0.0189	0.1558	4.1916	32	0.0037
common	5	total_vs_shared	0.4513	0.2825	0.6034	19.0429	30	0.0226
common	7	channel_increment	0.0968	0.0198	0.1745	4.8693	31	0.0096
common	7	total_vs_shared	0.4790	0.2960	0.6415	20.2102	29	0.0488
common	9	channel_increment	0.0897	0.0209	0.1577	4.4660	29	0.0488
common	9	total_vs_shared	0.4515	0.2642	0.6188	19.0508	27	0.1742
common	11	channel_increment	0.0911	0.0239	0.1552	4.6080	26	0.2912
common	11	total_vs_shared	0.4836	0.2911	0.6565	20.4059	30	0.0226
deployment	1	channel_increment	0.0909	0.0339	0.1483	4.0690	27	0.1742
deployment	1	total_vs_shared	0.1929	-0.0223	0.3828	8.2626	29	0.0488
deployment	2	channel_increment	0.0577	0.0137	0.1015	2.7430	30	0.0226
deployment	2	total_vs_shared	0.3073	0.1324	0.4802	13.0640	28	0.0961
deployment	3	channel_increment	0.0736	0.0289	0.1160	3.6318	33	0.0013
deployment	3	total_vs_shared	0.3915	0.2190	0.5594	16.6970	28	0.0961
deployment	5	channel_increment	0.0531	0.0127	0.0964	2.6343	29	0.0488
deployment	5	total_vs_shared	0.4014	0.2433	0.5478	16.9804	32	0.0037
deployment	7	channel_increment	0.0660	0.0117	0.1171	3.3206	30	0.0226
deployment	7	total_vs_shared	0.4175	0.2389	0.5840	17.8523	28	0.0961
deployment	9	channel_increment	0.0608	0.0080	0.1125	3.0403	30	0.0226
deployment	9	total_vs_shared	0.4188	0.2368	0.5845	17.7641	27	0.1742
deployment	11	channel_increment	0.0833	0.0177	0.1476	4.2276	25	0.4514
deployment	11	total_vs_shared	0.4819	0.2879	0.6639	20.3445	28	0.0961

Supplementary Note 13: Food scores against observed responses

The personal response level shifts every food by the same amount and therefore cannot change any within-participant ordering. This identity was verified numerically before the comparisons below, so only the channel slopes can move an ordering.

Within-participant pairwise concordance with observed responses does not improve with calibration in either cohort: −0.0012 (95% CI, −0.0103 to 0.0075) in free living and +0.0008 (−0.0049 to 0.0064) in standardized meals. Ties are frequent because each standardized-meal participant contributes seven meals; 749 of 996 participants are exactly tied, and a sign test that fails to exclude them is severely misleading.

Against a population nutritional-quality score the model orders observed responses 22.41 percentage points better (95% CI, 21.03–23.77). The shared predictor alone already achieves 22.33 points of that margin, so composition modelling rather than personalization delivers it.

Of the total variance of the 44 by 1,226 score matrix, 69.6% lies between participants, 8.4% between foods and 22.0% in the person-by-food interaction. The mean absolute personal adjustment is 2.94 score points. Personalization changes rankings substantially: the median rank shift is 94 places and a median of one food survives in a participant's top ten, while agreement with the nutritional-quality reference falls slightly, from a median Spearman correlation of 0.433 to 0.397.

The library contains 1,226 items with FNDDS 2017-18 food codes expressed per 100 kcal across 66 nutrient fields and 12 food groups, with a nutritional-quality reference for every item. Dairy and beverages contribute only 4 and 11 items, so no group-level conclusion is drawn for them. Fibre is zero for 12.0% of items and fat for 0.9%.

cohort	comparison	mean a	mean b	difference	participants	better	worse	tied	sign p excluding ties	ci
CGMacros	full_vs_shared	0.7058	0.7070	-0.0012	44	20	21	3	1.0000	-0.0
PREDICT_UK	full_vs_shared	0.9259	0.9251	0.0008	996	125	122	749	0.8988	-0.0
PREDICT_UK	full_vs_food_compass	0.9259	0.7018	0.2241	996	748	67	181	0.0000	0.2
PREDICT_UK	shared_vs_food_compass	0.9251	0.7018	0.2233	996	768	88	140	0.0000	0.2

block	quantity	value
variance_decomposition	cells	53944
variance_decomposition	foods	1226
variance_decomposition	participants	44
variance_decomposition	food_share_pct	8.369848874163965
variance_decomposition	participant_share_pct	69.64700570341856
variance_decomposition	interaction_share_pct	21.98314542241749
variance_decomposition	mean_abs_adjustment_score_points	2.9367744134678775
variance_decomposition	sd_adjustment_score_points	6.752597576727614
rank_shift_median	spearman	0.8764288385791936
rank_shift_median	median_abs_rank_change	94.25
rank_shift_median	p90_abs_rank_change	287.0
rank_shift_median	top10_overlap	1.0
rank_shift_median	spearman_vs_fcs2_shared	0.4334703404451742
rank_shift_median	spearman_vs_fcs2_personal	0.3973609874337672
rank_shift_median	foods	1226
food_library	foods	1226
food_library	source	FNDDS 2017-18 food codes (food_id prefix
food_library	exposure_basis	per 100 kcal
food_library	food_groups	12
food_library	fcs2_available	1226
food_library	nutrient_fields	66
food_library	fibre_zero_pct	11.99021207177814
food_library	fat_zero_pct	0.897226753670473

food group	foods
10000_Beverages	11
1000_Grains	143
2000_Fruit	57
3000_Vegetables	374
4000_LegNut	69
5000_MPE	98
5800_Seafood	83
6000_Dairy	4
7000_Fats Oils	23

food group	foods
8000_Mixed	235
8600_SauceCondiment	21
9000_SavorySweet	108

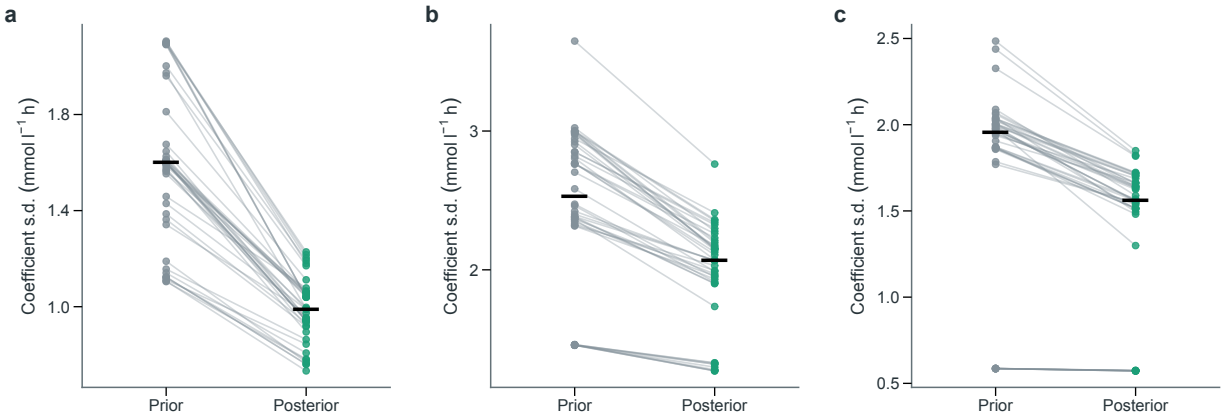
Supplementary Note 14: Cohort eligibility audit

Candidate additional cohorts held locally were audited for eligibility as adaptive validation data, which requires per-meal postprandial responses, microbial profiling and meal composition in the same participants. None qualifies. Cross-sectional microbiome resources carry no meal-level response. One resource carries per-meal responses but is recorded in the project provenance registry as simulation material and is excluded from all evidence. The two cohorts used in this work are therefore the complete set of eligible data.

resource	per meal postprandial	cgm	meal composition
CRA013939 (Zhang 2024 Adv Sci / CHNS)	False	False	False
predict_multi L2 predict1_real	True	True	False
external/american_gut	False	False	False
gmrepo / cmd sqlite snapshots	False	False	False
verification/zoe2025	False	False	False

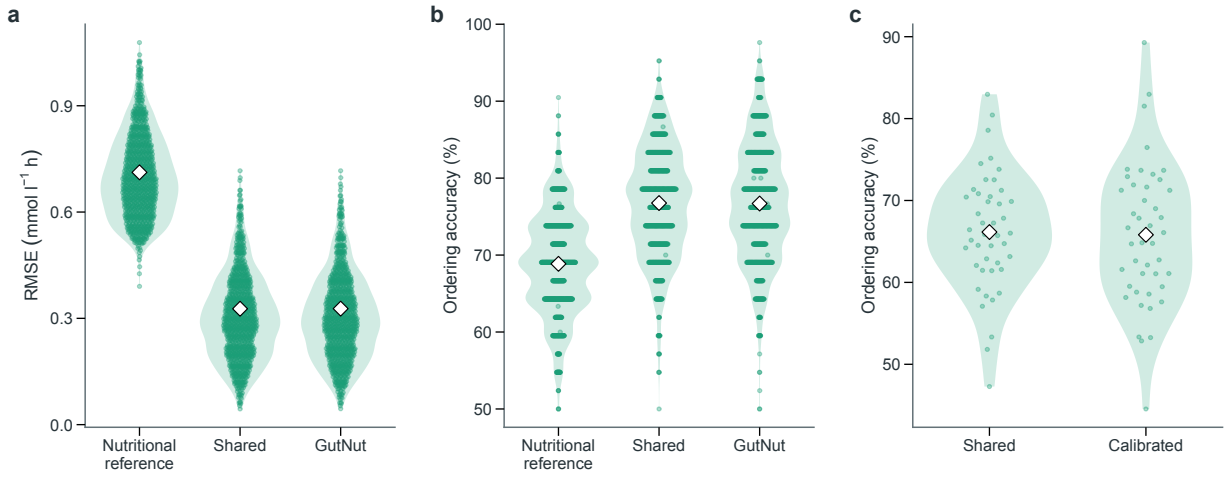
Supplementary Figures

Supplementary Fig. 1 | Personal measurements reduce coefficient uncertainty.



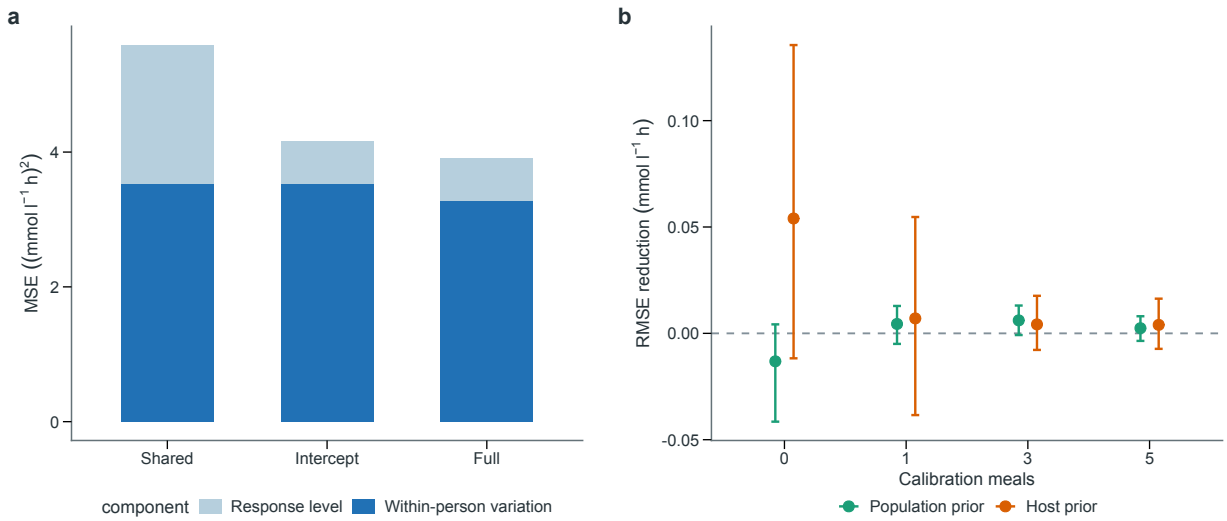
Prior and posterior coefficient standard deviations for the response level and the two channel slopes in 44 participants after three measured meals. Lines pair estimates from the same participant; black marks indicate medians. Median decreases are 36.4%, 19.5% and 17.5%. Standard deviations are conditional on the fitted model.

Supplementary Fig. 2 | Nutritional quality and response ordering.



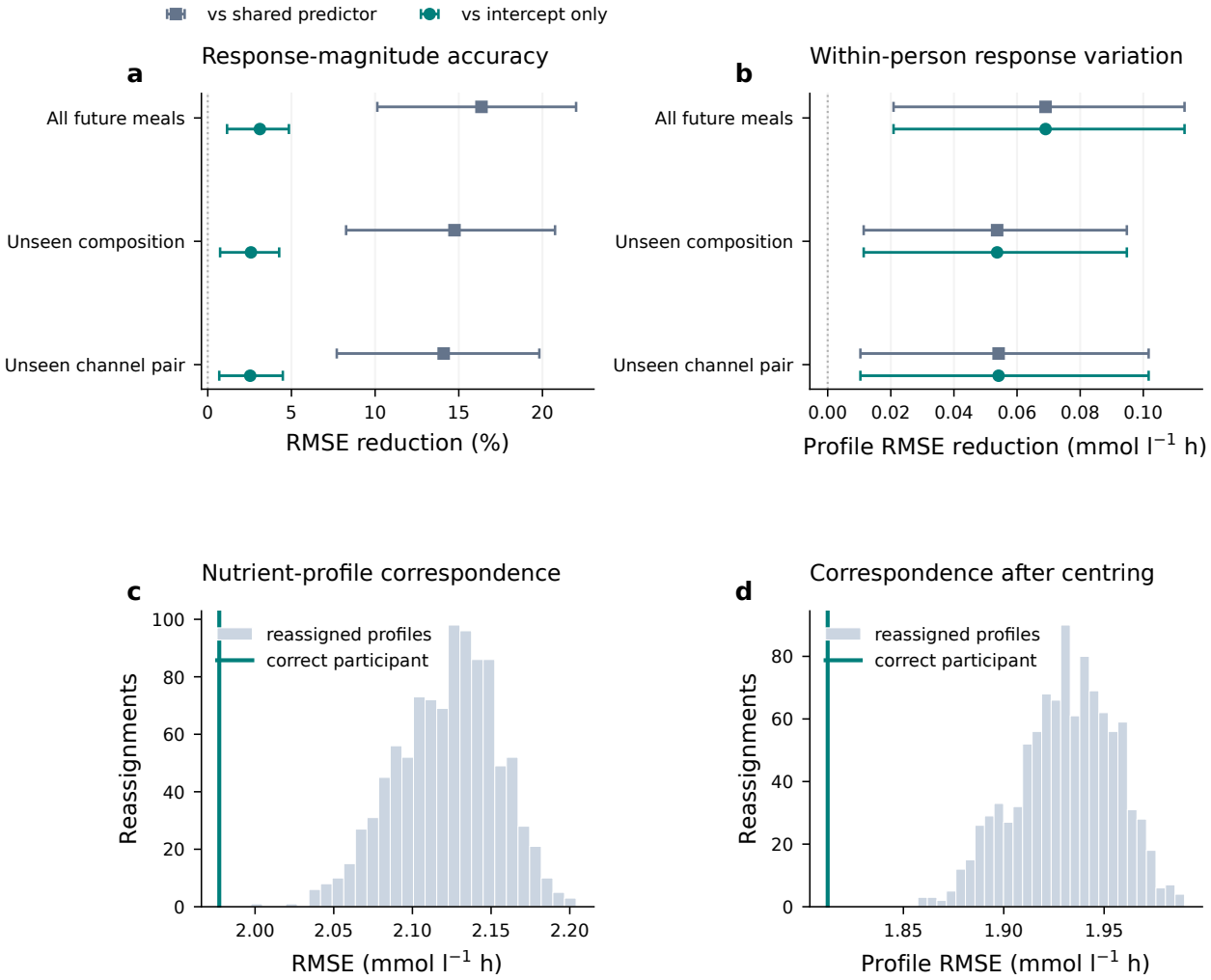
Participant-balanced error and within-participant ordering accuracy for an affine-calibrated nutritional score, the shared predictor and a static channel model. Ordering uses native score directions with half credit for ties.

Supplementary Fig. 3 | Error components and prior-information effects.



Participant-balanced mean squared error partitioned into within-person variation and squared mean error, and paired error reduction for the microbiome-informed prior relative to population and host priors across calibration histories.

Supplementary Fig. 4 | Personal nutrient information beyond calibration compositions.



Relative error reduction for all later meals, for compositions absent from personal calibration and for channel exposure pairs absent from personal calibration, together with the distribution of 1,000 joint reassignments of the channel slopes between participants within the same outer fold.